

Distortion cancellation of frequency converted pulses with simple linear signal processing and application to frequency modulation to amplitude modulation conversion in high power lasers

Sébastien Vidal,¹ Jacques Luce,¹ Steve Hocquet,¹ Claude Gouédard,²
Pierre Calvet,¹ and Denis Penninckx^{1,*}

¹CEA, DAM, CESTA, BP No. 2, F-33114 Le Barp, France

²CEA, DAM, DIF, Bruyères-le-Châtel, F-91297 Arpajon Cedex, France

*Corresponding author: denis.penninckx@cea.fr

Received 16 April 2012; revised 11 June 2012; accepted 12 July 2012;
posted 13 July 2012 (Doc. ID 166862); published 13 August 2012

It is known that a linear filter may be easily compensated with its inverse transfer function. However, it was shown that this approach could also be valid even for such a complex nonlinear system as frequency conversion. As a matter of fact, it is possible to at least partly precompensate for distortions occurring within, or even downstream from, frequency conversion crystals with a simple linear optical filter set upstream. In this paper, we give the theoretical background and derive the optimum precompensation filter from simple analytical formulas even in the case of saturation. We first show the relevance of our approach for Gaussian pulses: the pulse may be short or not and chirped or not, and the same linear precompensation filter may be used as long as saturation is not reached. We then study the case of phase-modulated pulses, as can be found on high power lasers such as lasers for fusion. We show that previous experimental results are in perfect agreement with these calculations. Finally, justified by our simple analytical formulas, we present a rigorous parametrical study giving the distortion reduction for any second and third harmonic generation system in the case of phase-modulated pulses. © 2012 Optical Society of America

OCIS codes: 190.2620, 070.1170.

1. Introduction

Megajoule-class lasers such as the Laser MegaJoule (LMJ) in France need to convert the amplified IR pulse into UV in order to enhance the coupling efficiency of the laser light with the target. To mitigate nonlinear issues (most likely stimulated Brillouin scattering) during propagation, the laser pulse has to be phase modulated at the beginning, in the fibered part of the laser [1,2]. Ideally, pure phase modulations should not affect the temporal shape of the pulse. However, any spectral filtering induces distortions called phase or frequency modulation (FM) to

amplitude modulation (AM) conversion [3,4]. This FM–AM conversion needs to be reduced, because it adversely affects the laser performance and thus can prevent fusion ignition and damage the optics. In the case of a linear transfer function, like chromatic dispersion in the fibered part of the laser or gain narrowing during amplification, it is relatively easy to add the inverse transfer function to cancel FM–AM conversion. However, for nonlinear filtering such as that due to the spectral acceptance of frequency conversion systems (FCSs) [5–7], signal processing is more complex. Moreover, it is much easier to add a compensation system when the optical power is low, because losses can be easily compensated at this stage. Hence, one would like to compensate distortions before amplification and frequency

conversion. We have previously shown both numerically and experimentally that this technique may be very efficient [8–11]. The goal of this paper is to provide a framework for understanding the efficiency and the limits of this technique.

In Section 2, this paper aims at explaining this phenomenon and we derive the optimum features of the precompensation filter from very simple analytical formulas both for Gaussian and phase-modulated pulses. In Section 3, we show the compliance of experimental results previously obtained. Finally, in Section 4 we will present rigorous parametrical studies in second-harmonic generation (SHG) and third-harmonic generation (THG), justified by the formalism developed earlier in the paper. These studies permit us to precisely quantify the nonlinear FM–AM conversion and its optimum compensation with simple parameters whatever the configuration of the frequency conversion process.

2. Analytical and Numerical Analysis in SHG

A. Formalism

For the sake of clarity, we will first consider the general case of SHG under the plane wave approximation without depletion of the fundamental wave. According to [12], the output converted electric field $E_2(\omega)$ at 2ω is proportional to $C(\omega) \times H(\omega)$, where $C(\omega) = \int_{-\infty}^{+\infty} E_1(\omega - \omega') E_1(\omega') d\omega'$, $H(\omega)$ is basically a “sinc” function, and $E_1(\omega)$ is the incident electric field at 1ω (ω is the frequency). $C(\omega)$ is the optical spectrum of the converted signal assuming perfect phase matching over the whole spectrum bandwidth; it can be seen as the ideal frequency conversion, and $H(\omega)$ can be seen as the distortions occurring after this ideal conversion.

Note that this formalism can also be applied to any linear transfer function occurring downstream from the frequency conversion. For instance in the LMJ case, the FCS is composed of two nonlinear crystals surrounded by two gratings [3]. The upstream grating is used to increase the spectral acceptance of the crystals by directing each wavelength to a direction in which the conversion efficiency is higher [10,13]. The nonlinear crystals induce a filtering in amplitude, while the gratings induce a filtering in phase. Thus, to take into account the distortions induced by the gratings, $H(\omega)$ becomes a complex transfer function [10].

Now, the aim is to find a function $H_{\text{comp}}(\omega)$ applied on the fundamental wave such that the impact of $H(\omega)$ is canceled:

$$[E_1(\omega)H_{\text{comp}}(\omega) \otimes E_1(\omega)H_{\text{comp}}(\omega)] \times H(\omega) = C(\omega), \quad (1)$$

where $\otimes$ means the convolution product.

B. Gaussian Pulses

It is important to note that there is no solution to Eq. (1) in the general case. However, the product and the convolution of Gaussian functions is a Gaussian

function as well. Hence, if $E_1(\omega)$ and $H(\omega)$ are Gaussian functions, even complex, their Fourier transforms are also Gaussian, and thus Eq. (1) has an analytic solution. As a matter of fact, writing $H(\omega) = \exp(-\sigma\omega^2)$ yields $H_{\text{comp}}(\omega) = \exp(-\sigma_{\text{comp}}\omega^2) = \exp(2\sigma\omega^2)$, where σ and σ_{comp} are complex parameters. Note that this solution is valid for any $E_1(\omega)$ as long as it is a Gaussian function; the pulse may be long or short, chirped or not. All this is confirmed by numerical simulations [cf. Figs. 1(a) and 1(b)]. The solid curve is the ideal converted signal. The dashed curve shows the distorted signal without compensation. The large dots show the same signal when a Gaussian precompensation is applied at 1ω with $\sigma_{\text{comp}} = -2\sigma$.

This analysis not only is theoretical but also has practical consequences because the basic “sinc” function due to frequency conversion can be approximated by a Gaussian function close to its peak [cf. Fig. 1(d)], and thus the precompensation with an inverse Gaussian function is almost perfect [cf. Fig. 1(c)]. Moreover, chromatic dispersion is an (imaginary) Gaussian function as well. Hence, a chromatic dispersion, a bell-shaped filtering, or even both occurring at 2ω may therefore be precompensated at 1ω for any Gaussian pulse.

Note that this model is valid for any frequency conversion order; for instance, $H_{\text{comp}}(\omega) = \exp(3\sigma\omega^2)$ in THG and so on.

C. Phase-Modulated Laser Pulses

Now, let us consider a typical LMJ pulse, which is much different from a complex Gaussian pulse: a sinusoidal phase modulation $\varphi(t) = m_{1\omega} \sin(2\pi f_m t)$, with $m_{1\omega}$ the modulation index and f_m the frequency modulation, is applied to the few-nanosecond pulse in order to create a $\Delta F_{1\omega} = 2 \cdot (m_{1\omega} + 1) \cdot f_m$ spectrum bandwidth composed of uniformly separated f_m -spaced Dirac peaks. Since the spectrum width is determined by the phase modulation and not by the pulse shape, we can consider in a first approximation, a signal given in the temporal domain by $E_1(t) = \exp[i\varphi(t)]$, neglecting the pulse shape.

First, we consider a Gaussian amplitude filtering (σ is real). Results of the numerical simulations are given in Fig. 2(a), which shows the intensity of $E_2(t)$ for one period of $E_1(t)$, where f_m is arbitrarily chosen to be equal to 5 GHz and $m_{1\omega} = 6$. The solid curve is the ideal constant signal after frequency conversion. The dashed curve shows the distorted signal without compensation. The large dots show the same signal when a Gaussian precompensation is applied at 1ω with $\sigma_{\text{comp}} = -2\sigma$, as is the case for the Gaussian pulses. Although the signal is not Gaussian at all, the same precompensation filter almost cancels the distortions. Now we consider phase filtering (σ is imaginary). Results are given in Fig. 2(b). Once again, distortions are tremendously reduced, but in this case $\sigma_{\text{comp}} = -\sigma$ and not -2σ .

To understand these differences, we should consider the time domain. To evaluate the FM–AM

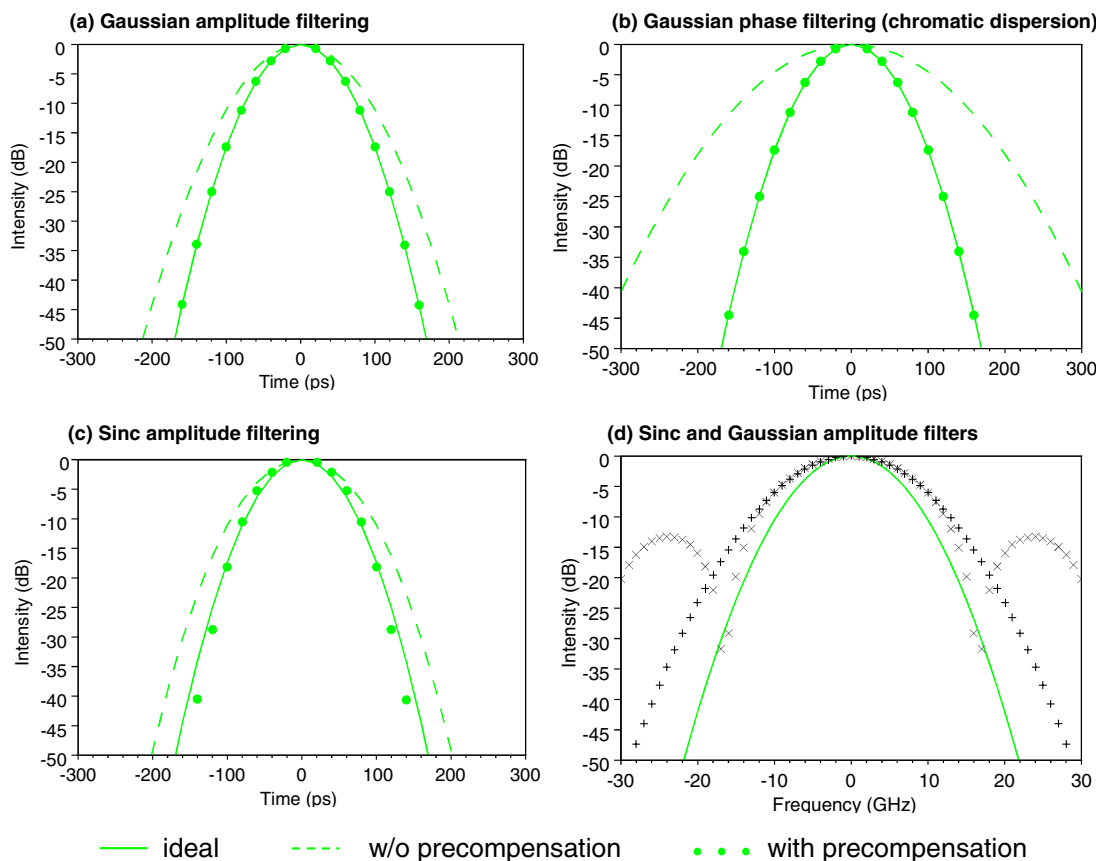


Fig. 1. (Color online) (a) Ideal Gaussian pulse filtered at 2ω by a Gaussian amplitude filter with $\sigma = 7 \cdot 10^{-21}$ international system of units (SI) without and with precompensation at 1ω ($\sigma_{\text{comp}} = -2\sigma$), (b) ideal Gaussian pulse filtered at 2ω by a Gaussian phase filter ($\sigma = i 2 \cdot 10^{-20}$ SI) without and with precompensation at 1ω ($\sigma_{\text{comp}} = -2\sigma$), (c) ideal Gaussian pulse filtered at 2ω by a sinc amplitude filter without and with precompensation by a Gaussian filter at 1ω , (d) spectrum of the ideal Gaussian pulse and filtering functions (sinc filtering is represented by the x 's and Gaussian filtering by the plus symbols).

conversion, the distortion criterion α , equal to twice the temporal contrast, is commonly used: $\alpha = 2 \cdot [(I_{\text{max}} - I_{\text{min}}) / (I_{\text{max}} + I_{\text{min}})]$, where I_{max} and I_{min} represent the maximum and minimum of pulse intensities, respectively. A phase filtering at 1ω (precompensation) generates FM-AM conversion proportional to $\alpha \approx \sigma_{\text{comp}} \cdot m_{1\omega} \cdot f_m^2$ [14]. Due to SHG, the AM is doubled because $I_2 = I_1^2 \cdot (1 + \text{AM})^2 \approx I_1^2 \cdot (1 + 2 \cdot \text{AM})$, where AM is a small amplitude modulation, I_1 (respectively I_2) represents the intensity of the fundamental wave (respectively the harmonic wave). Hence, since $m_{2\omega} = 2 \cdot m_{1\omega}$, the distortions are canceled if the distortions before and after compensation are opposite, i.e., $2 \cdot \sigma_{\text{comp}} \cdot m_{1\omega} \cdot f_m^2 = -\sigma \cdot (2 \cdot m_{1\omega}) \cdot f_m^2$, giving $\sigma_{\text{comp}} = -\sigma$.

Moreover, note that with precompensation, and without depletion of the fundamental wave, the AMs preinduced at 1ω are half the ones induced at 2ω : they are reduced and have less energetic photons and thus are much less dangerous for optics.

In amplitude, the FM-AM conversion is proportional to $\alpha \approx \sigma_{\text{comp}} \cdot m_{1\omega}^2 \cdot f_m^2$ [14]. Thus, $2 \cdot \sigma_{\text{comp}} \cdot m_{1\omega}^2 \cdot f_m^2 = -\sigma \cdot (2 \cdot m_{1\omega})^2 \cdot f_m^2$, yielding $\sigma_{\text{comp}} = -2 \cdot \sigma$. We can thus explain the difference between phase and amplitude filtering in the case of phase-modulated signals.

Once again, as the basic “sinc” function can be approximated by a Gaussian function [cf. Fig. 2(d)], the precompensation of the FM-AM conversion is almost perfect with a simple inverse Gaussian function [cf. Fig. 2(c)].

Note that the average shape of the pulse should be perfectly controlled for a fusion laser. Our technique reduces the somewhat higher frequencies due to FM-AM conversion. Moreover, this filtering technique does not affect the amplitude pulse shaping requirements for laser fusion.

D. Linear Precompensation near the Saturation of the Frequency Conversion Process

Up to now, this analysis seemed valid only when we could neglect the depletion of the fundamental wave, i.e., when the input intensity $I_1 \leq I_c$, where $I_c = P_c / L^2$ represents the critical intensity of the frequency conversion process, with L the crystal length and P_c the critical power of the crystal, defined in SHG by $P_c = \epsilon_0 c^3 n_1^2 n_2 / 4 \omega_0^2 \chi^{(2)2}$. In this expression, ϵ_0 is the vacuum permittivity, c is the velocity of light in the vacuum, $\chi^{(2)}$ is the second-order nonlinear susceptibility, and n_1 (respectively n_2) is the refractive index at ω_0 (respectively $2\omega_0$). Note that for $I_1 = I_c$, the conversion efficiency for the central

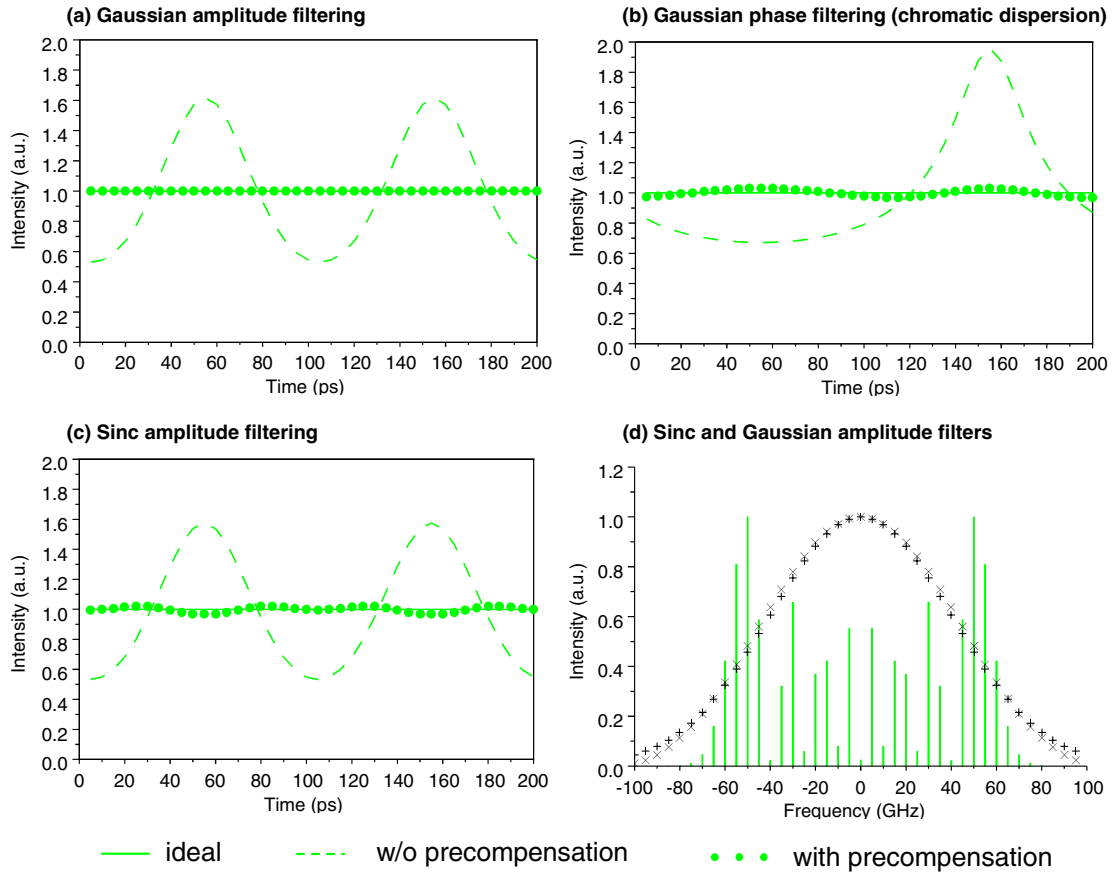


Fig. 2. (Color online) (a) Ideal phase-modulated constant signal ($f_m = 5$ GHz and $m_{1\omega} = 6$) filtered at 2ω by a Gaussian amplitude filter with $\sigma = 2 \cdot 10^{22}$ SI without and with precompensation at 1ω ($\sigma_{\text{comp}} = -2\sigma$), in the nondepleted regime, (b) same ideal phase-modulated constant signal filtered at 2ω by a Gaussian phase filter ($\sigma = i 8 \cdot 10^{-22}$ SI) without and with precompensation at 1ω ($\sigma_{\text{comp}} = -\sigma$), in the nondepleted regime, (c) same ideal phase-modulated constant signal filtered at 2ω by a sinc amplitude filter without and with precompensation by a Gaussian filter at 1ω , (d) spectrum of the ideal phase-modulated constant signal and filtering functions (sinc filtering is represented by the x 's and Gaussian filtering by the plus symbols).

wavelength (where we supposed $\Delta k = 0$) is around 55%. However, we show in the following that we can extend our approach near the saturation of the frequency conversion process when $I_1 > I_c$.

Near saturation, the transfer function becomes highly nonlinear; it is no longer a simple “sinc” function. Nevertheless, we can always approximate it by a Gaussian function with a bandwidth depending on I_1 , and we can search an average linear inverse Gaussian function to compensate for it. We can extend the previous simple model to this more complicated configuration, because near the saturation $I_2 \approx I_{2,0}(1 + \beta \cdot \text{AM})$, where $I_{2,0}$ is the mean intensity at 2ω and the parameter β depends on I_1/I_c and is given in Fig. 3(a).

As in Section 2.C, we get $\sigma_{\text{comp}} = -2^2\sigma/\beta$ in SHG and thus at the N th order of frequency conversion ($N = 2$ in SHG, 3 in THG, and so forth):

$$-\frac{\beta \cdot \sigma_{\text{comp}}}{\sigma} = N^2, \quad (2)$$

which can also be written as

$$\frac{\beta \cdot \gamma^2}{\gamma_{\text{comp}}^2} = N^2, \quad (3)$$

where γ is the full width at half-maximum (FWHM) of the frequency transfer function in intensity (usually the spectral acceptance of the crystal), and thus $\gamma^2 \propto 1/\sigma$. Equations (2) and (3) are very simple but, as we shall see, are valid over a very wide range.

To illustrate and validate this expression, we performed simulations in a 25 mm thick Type II potassium titanyl phosphate (KTP) crystal. In order to be representative of a previous experiment [9], f_m is chosen to be equal to 2 GHz and $m_{1\omega} = 7$. Results are given in Table 1 and are fully compliant with Eq. (3). α_{in} and α_{comp} are the FM–AM conversion factor, respectively, obtained without and with precompensation.

For a phase filtering at 1ω when $I_1 > I_c$, the AM is now multiplied by β , whereas $m_{2\omega}$ is still equal to $2 \cdot m_{1\omega}$ in SHG, leading finally to $\sigma_{\text{comp}} = -2\sigma/\beta$, and thus at the N th order of frequency conversion,

$$-\frac{\beta \cdot \sigma_{\text{comp}}}{\sigma} = N, \quad (4)$$

which can also be written as

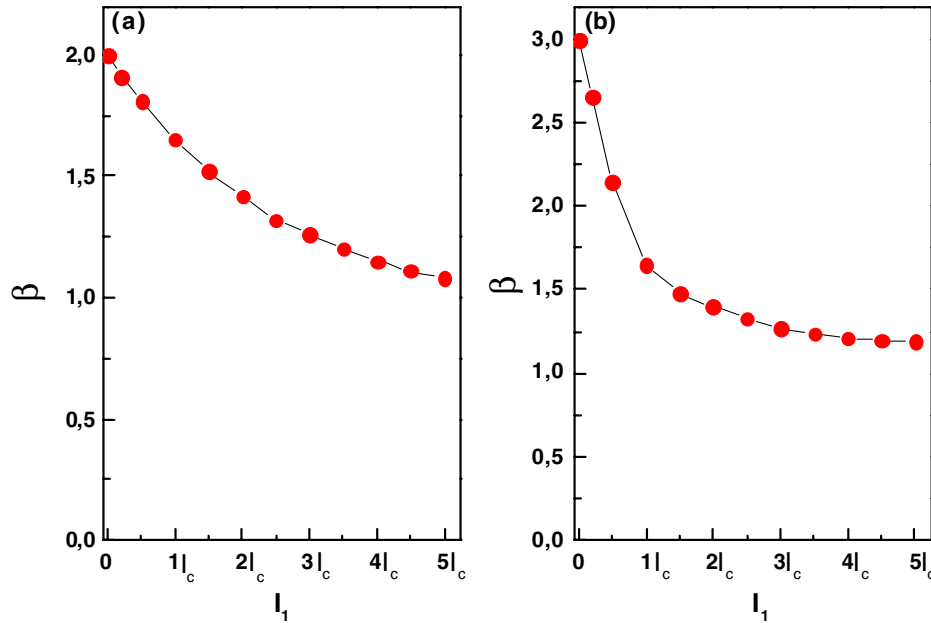


Fig. 3. (Color online) Evolution of the parameter β with intensity, expressing the saturation of the frequency conversion, in (a) SHG and (b) THG processes.

$$-\frac{\beta \cdot \varphi''_{\text{comp}}}{\varphi''_{2\omega}} = N, \quad (5)$$

where $\varphi''_{2\omega}$ is the usual way to write chromatic dispersion related to σ by $\sigma = i\varphi''_{2\omega}/2$. To illustrate and validate this expression, we performed simulations similar to the previous case in a 25 mm thick KTP crystal, but the amplitude filtering due to γ was canceled and we added phase filtering at 2ω such that $\sigma = i\varphi''_{2\omega}/2$. Results are given in Table 2 and are in agreement with Eq. (5).

In the following section, we show that previously obtained experimental results are also in perfect agreement with the formalism developed here.

3. Experimental Demonstrations in SHG

The laser system is composed of a fiber laser delivering square laser pulses at $\lambda_0 = 1053$ nm, an electro-optic phase modulator (EOM), and a regenerative

amplifier. At the output of the regenerative amplifier, we obtain a 2 mm collimated Gaussian beam with around 40 MW/cm² intensity. It should be noted that the weak FM-AM conversion due to gain narrowing in the regenerative amplifier is perfectly precompensated.

A. Nonlinear Filtering in Amplitude and in Phase

In the first experiment [10], we considered filtering both in amplitude and in phase due to a KTP crystal surrounded by two plane gratings. In this case, $I_1/I_c = 0.5$, $\gamma = 415$ GHz, and $\varphi''_{2\omega} = 1 \cdot 10^8$ fs². The FM was generated with the following parameters: $m_{1\omega} = 8$ and $f_m = 10$ GHz, leading to an ideal pulse bandwidth equal to $\Delta F_{2\omega} = 2 \cdot (2 \cdot m_{1\omega} + 1)f_m = 340$ GHz.

Without precompensation, the modulation factor α_{in} reaches around 105%. According to Eqs. (3) and (5), γ_{comp} should be equal to 293 GHz and φ''_{comp}

Table 1. Simulations Performed for Amplitude Filtering at 2ω

	β	γ (GHz)	α_{in} (%)	γ_{comp} (GHz)	α_{comp} (%)	$\beta \cdot \gamma^2 / \gamma_{\text{comp}}^2$
$I_1 = I_c/10$	2	80	34	56.5	≈ 0	4.0 ± 0.2
$I_1 = I_c$	1.7	77	38	49	≈ 0	4.2 ± 0.2
$I_1 = 2 \cdot I_c$	1.45	71	41	43	≈ 2	4.0 ± 0.2
$I_1 = 5 \cdot I_c$	1.05	59	55	29	≈ 16	4.3 ± 0.2

Table 2. Simulations Performed for Phase Filtering at 2ω

	β	$\varphi''_{2\omega} (\times 10^8 \text{ fs}^2)$	α_{in} (%)	$\varphi''_{\text{comp}} (\times 10^8 \text{ fs}^2)$	α_{comp} (%)	$-\varphi''_{\text{comp}} \beta / \varphi''_{2\omega}$
$I_1 = I_c/10$	2	-0.8	35	+0.8	≈ 0	2.0 ± 0.1
$I_1 = I_c$	1.7	-0.8	35	+0.94	≈ 0	2.0 ± 0.1
$I_1 = 2 \cdot I_c$	1.45	-0.8	35	+1.1	≈ 1	2.0 ± 0.1
$I_1 = 5 \cdot I_c$	1.05	-0.8	35	+1.5	≈ 14	2.0 ± 0.1

to $-1 \cdot 10^8 \text{ fs}^2$. For practical reasons, the inverse transfer functions in amplitude and in phase were generated with a two-wave interferometer [9] and a 320 m long polarization maintaining fiber (PMF), respectively. The fiber length effectively corresponded to $\phi''_{\text{comp}} = -1 \cdot 10^8 \text{ fs}^2$. Also, up to the second order in frequency, the sinusoidal transfer function of our amplitude filter could be approximated by an inverse Gaussian function with $\gamma_{\text{comp}} = 294 \text{ GHz}$. Thanks to this all-fiber system, $\alpha_{\text{comp}} = 11\%$. Numerical simulations show that these values were correct and α_{comp} could even be reduced with an inverse Gaussian filter instead of the two-wave interferometer. It could be designed with a fiber Bragg grating.

B. Nonlinear Filtering in Amplitude near the Saturation

In the second experiment [9], we showed the impact of frequency conversion saturation. We only considered amplitude filtering due to a 25 mm thick KTP crystal. We performed this experiment beyond the saturation of the SHG process, and thus γ highly depended on the input intensity. As a matter of fact, γ varied from 80 to 30 GHz when I_1 increased from 0 to 40 MW/cm². The FM was generated with the following parameters: $m_{1\omega} = 7$ and $f_m = 2 \text{ GHz}$, leading to an ideal pulse bandwidth equal to $\Delta F_{2\omega} = 60 \text{ GHz}$. With these values, the FM-AM conversion experimentally increased from $\alpha_{\text{in}} \approx 33\%$ to 130% with the intensity [9]. Once again, the inverse transfer function was experimentally generated with a two-wave interferometer. Its parameters were chosen to compensate FM-AM conversion at 10 MW/cm², for which $\beta = 1.2$ and $\gamma = 70 \text{ GHz}$. According to Eq. (3), γ_{comp} should be equal to 38.3 GHz. Experimentally, with $\gamma_{\text{comp}} = 42 \text{ GHz}$, α_{in} was reduced from 50% to $\alpha_{\text{comp}} = 8\%$, in agreement with numerical simulations. Once again, α_{comp} could even be reduced with an inverse Gaussian filter instead of the two-wave interferometer.

In the following section, we present rigorous parametrical studies of different frequency conversion processes.

4. Parametrical Studies

Since for amplitude filtering, $\alpha_{\text{in}} \propto \sigma N^2 m_{1\omega}^2 f_m^2 \propto \sigma \Delta F_{N\omega}^2$ [14], α_{in} only depends on $\Delta F_{N\omega}/\gamma$ [with $\Delta F_{N\omega} = N \cdot 2 \cdot (m_{1\omega} + 1) \cdot f_m$]. Moreover, according to Eq. (4), γ_{comp} only depends on γ and β (or, in other words, I_1/I_c). Hence, α_{comp} should only depend on $\Delta F_{N\omega}/\gamma$ and I_1/I_c .

We can thus perform a parametrical study in the case of SHG and THG with two parameters, $\Delta F_{N\omega}/\gamma$ and I_1/I_c .

A. Parametrical Study in SHG

Results in SHG are presented in Table 3. The back-conversion process (conversion efficiency reduction) is indicated in bold, and this gives the limit of our simple model.

Simulations were performed with the CEA's Miró propagation code [15]. We have considered different values of $m_{1\omega}$ and of f_m , different crystals, and different frequency conversion processes Type I or Type II: they are all valid in this table.

We can see that AM can be almost canceled for $I_1 \leq I_c$ whatever the ratio $\gamma/\Delta F_{2\omega}$, even below 1.

B. Parametrical Study in THG

We have developed a similar parametrical study in THG to be more representative of FCSs used in high power lasers. In this analysis, two nonlinear crystals are used to convert 1ω pulses to 2ω and 3ω successively, assuming that transfer functions are centered on the spectrum. I_c cannot be defined by the crystal, because we now have two crystals, so we consider a conversion efficiency of 55% at the central wavelength. Anyway, β is still related to I_1/I_c , as shown in Fig. 3(b). Once again we have verified that this table remains the same when the modulation parameters or the frequency process is changed.

C. Application to the LMJ Case

In this section, we will extend this analysis to the LMJ case. In the FCS, the IR beams (1ω) are converted into UV (3ω) thanks to a 14 mm thick Type I potassium dihydrogen phosphate (KDP) crystal and a 10 mm thick Type II potassium dideuterium phosphate (DKDP) crystal. An upstream plane grating enables us to widen the spectral acceptance of the crystals by directing each wavelength to a direction in which the conversion efficiency is higher [10,13]. We focus on nonlinear amplitude filtering due to the crystals, which is compliant with Subsection 4.B. Note that in the LMJ, the central wavelength is not perfectly phase matched in order to get the same number of photons at 1ω and 2ω , and thus the transfer function is slightly shifted by -17 GHz , but we will see that our parametrical analysis is still valid. In this configuration, $I_c = 0.8 \text{ GW/cm}^2$. In the LMJ, two phase modulations are applied: one avoids

Table 3. Parametrical Study in SHG^a

$I_1 = I_c/10$		$I_1 = I_c$		$I_1 = 2 \cdot I_c$		$I_1 = 5 \cdot I_c$	
$\gamma(I_1) = 3\Delta F_{2\omega}$	$\alpha_{\text{in}} = 5\%$ $\alpha_{\text{comp}} = 0\%$	$\alpha_{\text{in}} = 5\%$ $\alpha_{\text{comp}} = 0\%$	$\alpha_{\text{in}} = 5\%$ $\alpha_{\text{comp}} = 0\%$	$\alpha_{\text{in}} = 5\%$ $\alpha_{\text{comp}} = 0\%$	$\alpha_{\text{in}} = 5\%$ $\alpha_{\text{comp}} = 0\%$	$\alpha_{\text{in}} = 5\%$ $\alpha_{\text{comp}} = 0\%$	$\alpha_{\text{in}} = 5\%$ $\alpha_{\text{comp}} = 0\%$
$\gamma(I_1) = 2\Delta F_{2\omega}$	$\alpha_{\text{in}} = 13\%$ $\alpha_{\text{comp}} = 0\%$	$\alpha_{\text{in}} = 13\%$ $\alpha_{\text{comp}} = 0\%$	$\alpha_{\text{in}} = 13\%$ $\alpha_{\text{comp}} = 0\%$	$\alpha_{\text{in}} = 13\%$ $\alpha_{\text{comp}} = 0\%$	$\alpha_{\text{in}} = 13\%$ $\alpha_{\text{comp}} = 0\%$	$\alpha_{\text{in}} = 13\%$ $\alpha_{\text{comp}} = 0\%$	$\alpha_{\text{in}} = 13\%$ $\alpha_{\text{comp}} = 0\%$
$\gamma(I_1) = 1.5\Delta F_{2\omega}$	$\alpha_{\text{in}} = 23\%$ $\alpha_{\text{comp}} = 1.5\%$	$\alpha_{\text{in}} = 23\%$ $\alpha_{\text{comp}} = 0.3\%$	$\alpha_{\text{in}} = 23\%$ $\alpha_{\text{comp}} = 0.3\%$	$\alpha_{\text{in}} = 23\%$ $\alpha_{\text{comp}} = 1\%$	$\alpha_{\text{in}} = 23\%$ $\alpha_{\text{comp}} = 1\%$	$\alpha_{\text{in}} = 23\%$ $\alpha_{\text{comp}} = 3\%$	$\alpha_{\text{in}} = 23\%$ $\alpha_{\text{comp}} = 3\%$
$\gamma(I_1) = 1\Delta F_{2\omega}$	$\alpha_{\text{in}} = 55\%$ $\alpha_{\text{comp}} = 1.5\%$	$\alpha_{\text{in}} = 55\%$ $\alpha_{\text{comp}} = 3\%$	$\alpha_{\text{in}} = 55\%$ $\alpha_{\text{comp}} = 3\%$	$\alpha_{\text{in}} = 55\%$ $\alpha_{\text{comp}} = 6\%$	$\alpha_{\text{in}} = 55\%$ $\alpha_{\text{comp}} = 6\%$	$\alpha_{\text{in}} = 53\%$ $\alpha_{\text{comp}} = 12\%$	$\alpha_{\text{in}} = 53\%$ $\alpha_{\text{comp}} = 12\%$
$\gamma(I_1) = 0.75\Delta F_{2\omega}$	$\alpha_{\text{in}} = 95\%$ $\alpha_{\text{comp}} = 4\%$	$\alpha_{\text{in}} = 95\%$ $\alpha_{\text{comp}} = 7\%$	$\alpha_{\text{in}} = 95\%$ $\alpha_{\text{comp}} = 7\%$	$\alpha_{\text{in}} = 92\%$ $\alpha_{\text{comp}} = 18\%$	$\alpha_{\text{in}} = 92\%$ $\alpha_{\text{comp}} = 18\%$	$\alpha_{\text{in}} = 88\%$ $\alpha_{\text{comp}} = 36\%$	$\alpha_{\text{in}} = 88\%$ $\alpha_{\text{comp}} = 36\%$

^a α_{in} and α_{comp} are the FM-AM conversion factors obtained without compensation and with an optimized inverse Gaussian precompensation function, respectively. The backconversion process is indicated in bold.

Table 4. Parametrical Study in THG^a

$I_1 = I_c/10$		$I_1 = I_c$		$I_1 = 1.5 \cdot I_c$		$I_1 = 2.5 I_c$	
$\gamma(I_1) = 3\Delta F_{3\omega}$	$\alpha_{in} = 5\%$ $\alpha_{comp} = 0\%$	$\alpha_{in} = 5\%$ $\alpha_{comp} = 0\%$	$\alpha_{in} = 5\%$ $\alpha_{comp} = 0.7\%$	$\alpha_{in} = 5\%$ $\alpha_{comp} = 1\%$	$\alpha_{in} = 5\%$ $\alpha_{comp} = 1\%$	$\alpha_{in} = 5\%$ $\alpha_{comp} = 1\%$	$\alpha_{in} = 5\%$ $\alpha_{comp} = 1\%$
$\gamma(I_1) = 2\Delta F_{3\omega}$	$\alpha_{in} = 12\%$ $\alpha_{comp} = 0\%$	$\alpha_{in} = 12\%$ $\alpha_{comp} = 0\%$	$\alpha_{in} = 12\%$ $\alpha_{comp} = 1\%$	$\alpha_{in} = 12\%$ $\alpha_{comp} = 1\%$	$\alpha_{in} = 12\%$ $\alpha_{comp} = 3\%$	$\alpha_{in} = 12\%$ $\alpha_{comp} = 3\%$	$\alpha_{in} = 12\%$ $\alpha_{comp} = 3\%$
$\gamma(I_1) = 1.5\Delta F_{3\omega}$	$\alpha_{in} = 23\%$ $\alpha_{comp} = 0.6\%$	$\alpha_{in} = 23\%$ $\alpha_{comp} = 1\%$	$\alpha_{in} = 23\%$ $\alpha_{comp} = 2\%$	$\alpha_{in} = 23\%$ $\alpha_{comp} = 2\%$	$\alpha_{in} = 23\%$ $\alpha_{comp} = 6\%$	$\alpha_{in} = 23\%$ $\alpha_{comp} = 6\%$	$\alpha_{in} = 23\%$ $\alpha_{comp} = 6\%$
$\gamma(I_1) = 1\Delta F_{3\omega}$	$\alpha_{in} = 57\%$ $\alpha_{comp} = 2.3\%$	$\alpha_{in} = 57\%$ $\alpha_{comp} = 7\%$	$\alpha_{in} = 57\%$ $\alpha_{comp} = 16\%$	$\alpha_{in} = 57\%$ $\alpha_{comp} = 16\%$	$\alpha_{in} = 57\%$ $\alpha_{comp} = 34\%$	$\alpha_{in} = 57\%$ $\alpha_{comp} = 34\%$	$\alpha_{in} = 57\%$ $\alpha_{comp} = 34\%$
$\gamma(I_1) = 0.75\Delta F_{3\omega}$	$\alpha_{in} = 105\%$ $\alpha_{comp} = 6.5\%$	$\alpha_{in} = 105\%$ $\alpha_{comp} = 20\%$	$\alpha_{in} = 105\%$ $\alpha_{comp} = 48\%$	$\alpha_{in} = 105\%$ $\alpha_{comp} = 48\%$	$\alpha_{in} = 102\%$ $\alpha_{comp} = 65\%$	$\alpha_{in} = 102\%$ $\alpha_{comp} = 65\%$	$\alpha_{in} = 102\%$ $\alpha_{comp} = 65\%$

^a α_{in} and α_{comp} are the FM-AM conversion factors obtained without compensation and with an optimized inverse Gaussian precompensation function, respectively. The backconversion process is indicated in bold.

stimulated Brillouin scattering ($m_{1\omega,B} = 7$ and $f_{m,B} = 2$ GHz), and the other enables us to smooth the speckle pattern ($m_{1\omega,S} = 5$ and $f_{m,S} = 14.25$ GHz). The spectrum is the convolution of both, and the spectrum bandwidth at 3ω is thus about $\Delta F_{3\omega} \approx 600$ GHz. We have performed simulations of frequency conversion for two input intensities, $I_1 = 0.2$ GW/cm² and 2 GW/cm².

At a low intensity (0.2 GW/cm²), the spectral acceptance of the FCS is around $\gamma = 1220$ GHz, leading to an FM-AM conversion factor around $\alpha_{in} = 13\%$. AM is almost canceled ($\alpha_{comp} = 1\%$), in agreement with Table 4. Numerically, we find $\gamma_{comp} = 670$ GHz, very close to the theoretical value given by Eq. (3): 668 GHz.

At high intensity (2 GW/cm²), $\gamma = 900$ GHz, leading to $\alpha_{in} = 23\%$. AM can be divided by 3 ($\alpha_{comp} = 7\%$) in agreement with Table 4. Numerically, we find $\gamma_{comp} = 335$ GHz, very close to the theoretical value given by Eq. (3): 342 GHz.

This study shows that it is possible to use linear transfer functions to compensate part of the intensity modulation due to the FCS for any considered intensity. The drawback of this technique is that reduction is maximized for a unique intensity, and so a pulse (such as LMJ) with a particular temporal intensity shape is not compensated at any time; choices have to be made considering more disadvantageous intensity modulations. For laser plasma interaction, the first part of the ignition pulse at a low intensity has to be as stable as possible, so FM-AM conversion value reduction is more interesting. However, for laser damage, reduction of high intensity AM modulation should be more beneficial.

5. Conclusion

In summary, we have studied analytically, numerically, and experimentally the relevance of a linear optical filter set upstream from an FCS to precompensate for nonlinear distortions occurring during or downstream from frequency conversion.

First, we have shown that a chromatic dispersion, a bell-shaped filtering, or even both occurring after frequency conversion may be precompensated at 1ω . The compensation is even perfect if the pulse and the filters may be considered as Gaussian. In that case, the precompensation filter does not change whether the pulse is short or long, chirped or not.

Then, we have extended this analysis to the precompensation of FM-AM conversion due to FCSs, and we have proposed analytical expressions of the

optimized precompensation function (in phase and in amplitude), also applicable near the saturation of the process. We have shown that these analytical expressions are fully compliant with numerical simulations and experimental results.

Finally, suggested by the analytical expressions, we have developed parametrical studies in the case of amplitude filtering in SHG and THG, permitting us to precisely quantify the FM-AM conversion ratio and its compensation whatever the configuration of the frequency conversion process, the nonlinear crystals, and the phase modulation parameters.

References

1. J. R. Murray, J. R. Smith, R. B. Ehrlich, D. T. Karazys, C. E. Thompson, T. L. Weiland, and R. B. Wilcox, "Experimental observation and suppression of transverse stimulated Brillouin scattering in large optical components," *J. Opt. Soc. Am. B* **6**, 2402–2411 (1989).
2. J. Garnier, L. Videau, C. Gouédard, and A. Migus, "Statistical analysis for beam smoothing and some applications," *J. Opt. Soc. Am. A* **14**, 1928–1937 (1997).
3. S. Hocquet, D. Penninckx, E. Bordenave, C. Gouédard, and Y. Jaouën, "FM-to-AM conversion in high-power lasers," *Appl. Opt.* **47**, 3338–3349 (2008).
4. J. E. Rothenberg, D. F. Browning, and R. B. Wilcox, "Issue of FM-to-AM conversion on the National Ignition Facility," *Proc. SPIE* **3492**, 51–61 (1999).
5. Y. Yang, B. Feng, W. Han, W. Zheng, F. Li, and J. Tan, "Suppression of FM-to-AM conversion in third-harmonic generation at the retracing point of a crystal," *Opt. Lett.* **34**, 3848–3850 (2009).
6. Y. Chen, P. Yuan, L. Qian, H. Zhu, and D. Fan, "Numerical study on the efficient generation of 351 nm broadband pulses by frequency mixing of broadband and narrowband Nd:glass lasers," *Opt. Commun.* **283**, 2737–2741 (2010).
7. W. Wang, W. Han, F. Wang, J. Wang, L. Zhou, H. Jia, Y. Xiang, K. Li, F. Li, L. Wang, W. Zhong, X. Zhang, S. Zhao, and B. Feng, "Comparison of efficient third-harmonic generation between phase modulated broadband and narrowband lasers," *J. Opt. Soc. Am. B* **28**, 475–482 (2011).
8. S. Hocquet, G. Lacroix, and D. Penninckx, "Compensation of frequency modulation to amplitude modulation conversion in frequency conversion systems," *Appl. Opt.* **48**, 2515–2521 (2009).
9. S. Vidal, J. Luce, and D. Penninckx, "Experimental demonstration of linear precompensation of a nonlinear transfer function due to second-harmonic generation," *Opt. Lett.* **36**, 88–90 (2011).
10. S. Vidal, J. Luce, and D. Penninckx, "Compensation of phase-to-amplitude modulation conversion in a complete frequency conversion system with an all-fiber system," *Opt. Lett.* **36**, 3494–3496 (2011).
11. D. Penninckx, S. Hocquet, and C. Gouédard, "Dispositif de réduction des distorsions temporelles induites dans des impulsions lumineuses par un système convertisseur de fréquence optique non linéaire," French patent application 08-58954 (2008).

12. S. A. Akhmanov, V. A. Vysloukh, and A. S. Chirkin, *Optics of Femtosecond Laser Pulses* (American Institute of Physics, 1992).
13. R. W. Short and S. Skupsky, "Frequency conversion of broad-bandwidth laser light," *IEEE J. Quantum Electron.* **26**, 580–588 (1990).
14. D. Penninckx, N. Beck, J. F. Gleyze, and L. Videau, "Signal propagation over polarization-maintaining fibers: problem and solutions," *J. Lightwave Technol.* **24**, 4197–4207 (2006).
15. O. Morice, "Miró: complete modeling and software for pulse amplification and propagation in high-power lasers systems," *Opt. Eng.* **42**, 1530–1541 (2003).